

PRACTICE PROGRAM



Secure Exit

SIMATS Online Programming Lab
JAVA PROGRAMMING

 HARITHA K S
192366001RD

QUESTIONS 21 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int year = sc.nextInt();
8
9         if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
10            System.out.println("Leap Year");
11        else
12            System.out.println("Not a Leap Year");
13
14        sc.close();
15    }
16 }
```

OUTPUT

Leap Year

INPUT

2024



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QUESTIONS 21 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int a = 0, b = 1;
9
10        for (int i = 1; i <= n; i++) {
11            System.out.print(a + " ");
12            int c = a + b;
13            a = b;
14            b = c;
15        }
16
17        sc.close();
18    }
19 }
```

OUTPUT

0 1 1 2 3 5 8

INPUT

7



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QUESTIONS 21 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         long fact = 1;
9
10        for (int i = 1; i <= n; i++) {
11            fact *= i;
12        }
13
14        System.out.println(fact);
15
16        sc.close();
17    }
18 }
```

OUTPUT

120

INPUT

5



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QUESTIONS 21 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:
**

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         for (int i = 1; i <= 5; i++) {
4             for (int j = 1; j <= i; j++) {
5                 System.out.print("**");
6             }
7             System.out.println();
8         }
9     }
10 }
```

OUTPUT

**

INPUT

Your INPUT goes here!



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QUESTIONS 21 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int key = sc.nextInt();
14
15        int low = 0, high = n - 1;
16        boolean found = false;
17
18        while (low <= high) {
19            int mid = (low + high) / 2;
20
21            if (a[mid] == key) {
```

OUTPUT

Compilation Error:

INPUT

5
10 20 30 40 50
30



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QUESTIONS 21 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9
10        while (b != 0) {
11            int temp = b;
12            b = a % b;
13            a = temp;
14        }
15
16        System.out.println(a);
17
18        sc.close();
19    }
20 }
```

OUTPUT

12

INPUT

24
36



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QUESTIONS 21 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         boolean prime = true;
9
10        if (n <= 1) {
11            prime = false;
12        } else {
13            for (int i = 2; i <= n / 2; i++) {
14                if (n % i == 0) {
15                    prime = false;
16                    break;
17                }
18            }
19        }
20
21        if (prime)
```

OUTPUT

Prime

INPUT

13



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QUESTIONS 21 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        for (int i = 0; i < n - 1; i++) {
14            int min = i;
15            for (int j = i + 1; j < n; j++) {
16                if (a[j] < a[min])
17                    min = j;
18            }
19
20            int temp = a[i];
21            a[i] = a[min];
```

OUTPUT

11 12 22 25 64

INPUT

5
64 25 12 22 11



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QUESTIONS 21 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8
9         for (int i = 1; i <= 10; i++) {
10             System.out.println(n + " x " + i + " = " + (n * i));
11         }
12
13         sc.close();
14     }
15 }
```

OUTPUT

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

INPUT

5



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QUESTIONS 21 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     static int factorial(int n) {
6         if (n == 0 || n == 1)
7             return 1;
8         else
9             return n * factorial(n - 1);
10    }
11
12    public static void main(String[] args) {
13        Scanner sc = new Scanner(System.in);
14
15        int n = sc.nextInt();
16        System.out.println(factorial(n));
17
18        sc.close();
19    }
20 }
```

OUTPUT

120

INPUT

5



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QUESTIONS 21 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR

Java

Run

Save

```

1 public class Main {
2     public static void main(String[] args) {
3
4
5         for (int i = 1; i <= 5; i++) {
6             if (i == 5) {
7                 System.out.println("*****");
8             } else {
9                 System.out.println("");
10            }
11        }
12
13        System.out.println();
14
15        System.out.println("*****");
16        for (int i = 1; i <= 4; i++) {
17            System.out.println("  ");
18        }
19    }
20 }
21

```

OUTPUT

```

*
*
*
*
*****
*****
*
*
*

```

INPUT

Your INPUT goes here!



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QUESTIONS 21 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9
10        int max = (a > b) ? a : b;
11
12        while (true) {
13            if (max % a == 0 && max % b == 0) {
14                System.out.println(max);
15                break;
16            }
17            max++;
18        }
19
20        sc.close();
21    }
22 }

```

OUTPUT

```

36

```

INPUT

12
18



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QUESTIONS 21 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         String binary = "";
9
10        while (n > 0) {
11            binary = (n % 2) + binary;
12            n = n / 2;
13        }
14
15        System.out.println(binary);
16
17        sc.close();
18    }
19 }
```

OUTPUT

1010

INPUT

10



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QUESTIONS 21 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int binary = sc.nextInt();
8         int decimal = 0, base = 1;
9
10        while (binary > 0) {
11            int rem = binary % 10;
12            decimal = decimal + rem * base;
13            base = base * 2;
14            binary = binary / 10;
15        }
16
17        System.out.println(decimal);
18
19        sc.close();
20    }
21 }
```

OUTPUT

10

INPUT

1010



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QUESTIONS 21 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int sum = 0;
9
10        while (n > 0) {
11            sum = sum + (n % 10);
12            n = n / 10;
13        }
14
15        System.out.println(sum);
16
17        sc.close();
18    }
19 }
```

OUTPUT

10

INPUT

1234



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QUESTIONS 21 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int product = 1;
9
10        while (n > 0) {
11            product = product * (n % 10);
12            n = n / 10;
13        }
14
15        System.out.println(product);
16
17        sc.close();
18    }
19 }
```

OUTPUT

24

INPUT

1234



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QUESTIONS 21 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int product = 1;
9
10        while (n > 0) {
11            product = product * (n % 10);
12            n = n / 10;
13        }
14
15        System.out.println(product);
16
17        sc.close();
18    }
19 }
```

OUTPUT

24

INPUT

1234



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QUESTIONS 21 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int base = sc.nextInt();
8         int exponent = sc.nextInt();
9
10        int power = 1;
11
12        for (int i = 1; i <= exponent; i++) {
13            power = power * base;
14        }
15
16        System.out.println(power);
17
18        sc.close();
19    }
20 }
```

OUTPUT

32

INPUT

2
5

QUESTIONS 21 / 31 completed

Area Circle ▾

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         double radius = sc.nextDouble();
8         double area = 3.14 * radius * radius;
9
10        System.out.println(area);
11
12        sc.close();
13    }
14 }
```

OUTPUT

78.5

INPUT

5



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QUESTIONS 21 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         double radius = sc.nextDouble();
8         double volume = (4.0 / 3.0) * 3.14 * radius * radius * radius;
9
10        System.out.println(volume);
11
12        sc.close();
13    }
14 }
```

OUTPUT

```
113.03999999999998
```

INPUT

```
3
```



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QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         String str = sc.next();
8         boolean palindrome = true;
9
10        for (int i = 0; i < str.length() / 2; i++) {
11            if (str.charAt(i) != str.charAt(str.length() - 1 - i)) {
12                palindrome = false;
13                break;
14            }
15        }
16
17        if (palindrome)
18            System.out.println("Palindrome");
19        else
20            System.out.println("Not Palindrome");
21    }
22 }
```

OUTPUT

```
Palindrome
```

INPUT

```
madam
```



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QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int temp = n, sum = 0, digit;
9
10        while (temp > 0) {
11            digit = temp % 10;
12            sum += digit * digit * digit;
13            temp /= 10;
14        }
15
16        if (sum == n)
17            System.out.println("Armstrong");
18        else
19            System.out.println("Not Armstrong");
20    }
21 }
```

OUTPUT

Armstrong

INPUT

153



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QUESTIONS 31 / 31 completed

RT-num.Pattern

**RT-
NUM.PATTERN**

Write a Java program to print the following pattern using loops:

1
22
333
4444
55555

PROGRAM EDITOR

Java Run Save

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         for (int i = 1; i <= 5; i++) {
6             for (int j = 1; j <= i; j++) {
7                 System.out.print(i);
8             }
9             System.out.println();
10        }
11    }
12 }
```

OUTPUT

1
22
333
4444
55555

INPUT

Your INPUT goes here!



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QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int sum = 0;
9
10        for (int i = 1; i < n; i++) {
11            if (n % i == 0)
12                sum += i;
13        }
14
15        if (sum == n)
16            System.out.println("Perfect Number");
17        else
18            System.out.println("Not Perfect Number");
19    }
20 }
```

OUTPUT

Perfect Number

INPUT

6



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QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int max = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] > max)
17                max = a[i];
18        }
19
20        System.out.println(max);
21    }
22 }
```

OUTPUT

40

INPUT

5
10 25 8 40 15



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QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int largest = a[0], second = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] > largest) {
17                second = largest;
18                largest = a[i];
19            } else if (a[i] > second && a[i] != largest) {
20                second = a[i];
21            }
22        }
23    }
24 }
```

OUTPUT

30

INPUT

5
10 25 40 15 30



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QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int min = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] < min)
17                min = a[i];
18        }
19
20        System.out.println(min);
21    }
22 }
```

OUTPUT

8

INPUT

5
10 25 8 40 15



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QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int smallest = Integer.MAX_VALUE;
14        int second = Integer.MAX_VALUE;
15
16        for (int i = 0; i < n; i++) {
17            if (a[i] < smallest) {
18                second = smallest;
19                smallest = a[i];
20            } else if (a[i] < second && a[i] != smallest) {
21                second = a[i];
22            }
23        }
24    }
25 }
```

OUTPUT

10

INPUT

5
10 25 8 40 15



Secure Exit

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QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        for (int i = 0; i < n - 1; i++) {
14            for (int j = 0; j < n - 1 - i; j++) {
15                if (a[j] > a[j + 1]) {
16                    int temp = a[j];
17                    a[j] = a[j + 1];
18                    a[j + 1] = temp;
19                }
20            }
21        }
22    }
23 }
```

OUTPUT

12 22 25 34 64

INPUT

5
64 34 25 12 22

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        for (int i = 1; i < n; i++) {
14            int key = a[i];
15            int j = i - 1;
16
17            while (j >= 0 && a[j] > key) {
18                a[j + 1] = a[j];
19                j--;
20            }
21        }
```

OUTPUT

12 22 25 34 64

INPUT

5
64 34 25 12 22